Martín Cogo

Computer Vision Engineer | Real-Time Video Perception & Edge AI

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Summary

I build real-time computer vision systems on NVIDIA GPUs: GStreamer/DeepStream pipelines running TensorRT models on live RTSP cameras, with per-stage telemetry to measure and optimize latency. At MindColab I cut a live pipeline's latency from ~1.1 s to ~39 ms and authored 98% of the commits of a 12-component edge platform. Computer Science graduate (UNCUYO) with the highest GPA of the 2024 class. Looking for Computer Vision / Perception Engineer roles.

Experience

MindColab

Computer Vision Software Engineer (Part-time)
July 2025 – present
Real-time inference engine and edge platform for visual inspection of mining conveyor belts. It processes multiple live RTSP cameras with GStreamer/DeepStream and TensorRT, keeping frames in GPU memory (NVMM).

- **Latency**: Cut live pipeline latency from ~1.1 s to ~39 ms mean (~28×; 76 ms p99 against a 200 ms frame budget) and frame loss from 17.9% to 0.11% in steady state. I removed two hidden waits that my telemetry exposed: an encoder blocking upstream threads and publisher clock sync.
- **Per-stage telemetry**: Designed and built the engine's telemetry: 48 timing buckets per job, probes on every GStreamer element, GPU timings via CUDA events, mergeable log histograms (p50/p95/p99) in 5 s windows, and a REST API with GPU, CPU and RAM metrics.
- **Models**: Runs 2 TensorRT FP16 models per frame: DINOv2-small detects foreign objects (anomaly detection) and FastSAM-s segments rocks and measures their size. Compared Dinomaly and EfficientAD, and validated ONNX and TensorRT exports against the original PyTorch checkpoint.
- **24/7 operation**: Self-healing workers: 8 automatic restarts in ~1 h after upstream RTSP source drops, each back in 34–36 s with no manual action. Controlled restarts, config hot-reload through a custom DSL, and graceful shutdown.
- **Platform**: Main author (98% of 2,717 commits) of a 12-component edge platform: FastAPI microservices, ZMQ, Redis, a React SPA with live WebRTC video, and centralized logging (Fluent Bit + VictoriaLogs), deployed with k3s + Helm for NVIDIA Jetson.

Stack: Python, GStreamer, DeepStream, TensorRT, CUDA, PyTorch, ONNX, Anomalib, Ultralytics, OpenCV, FastAPI, ZMQ, Redis, React, Docker, k3s, Helm

Municipality of Godoy Cruz

Web Developer (Part-time)
June 2025 – Sept 2026
Maintained and extended the municipal debt and tax payment web application (GeneXus-generated Java).

- **Payment integrations**: Refactored integrations with 6 banking payment gateways (Banco Macro, Mercado Pago, ePagos, XNet, Pago Fácil, Supervielle) in the amount and transaction submission flow.
- **LLM chat**: Built an internal chat interface to query municipal regulations through semantic search and an LLM (view project).

Stack: Java, GeneXus, Tomcat, Nginx, React, Tailwind, Bun

Klari Inc.

Data Scientist (Internship, 200 hrs)
Jan 2025 – Feb 2025
Semantic analysis of conversations from an LLM-based chatbot, with reports on interaction quality.

- **NLP pipeline**: Conversation embeddings and unsupervised clustering to detect recurring topics.

Stack: Python, NumPy, Pandas, Scikit-learn, PyTorch, FastAPI, MongoDB

CICDa Research Group
Undergraduate Thesis Researcher
Jan 2024 – Jan 2025
Thesis on post-disaster building damage assessment from satellite imagery: a multitask Siamese U-Net that segments buildings and classifies damage into 4 levels from pre- and post-disaster images (code · thesis, in Spanish).

- **End-to-end pipeline**: Preprocessing, training, inference and post-processing on ~3,000 VHR satellite images from the xBD dataset, with a custom greedy sampling strategy to balance damage classes.
- **Results**: 0.80 F1 for building segmentation on the test split, with pixel- and object-level evaluation.
- **Compute**: Trained on an HPC cluster with NVIDIA Tesla V100 GPUs.

Stack: Python, PyTorch, Pandas, Scikit-learn, OpenCV, Linux, Git

Technical Skills

Programming Languages: Python, TypeScript/JavaScript, C/C++

Computer Vision & Inference: GStreamer, DeepStream, TensorRT, CUDA, ONNX, OpenCV

AI & ML: PyTorch, Ultralytics, Anomalib, Scikit-learn, Pandas, NumPy

Backend, REST APIs & Web: FastAPI, ZMQ, Nginx, React, Tailwind

Databases: Redis, SQLite, MongoDB

DevOps & Tools: Docker, k3s, Helm, Linux, Git

Education

National University of Cuyo
Bachelor of Computer Science, Undergraduate Degree (GPA 8.7/10)
2020 – 2024

Awards

- Highest GPA (8.7 on a ten-point scale) in the Computer Science program, Class of 2024.
- Faculty of Engineering honor roll (top 15 GPAs), UNCUYO, 2024.

Languages

- Spanish: Native.
- English: Upper-Intermediate (B2), Cambridge certified.